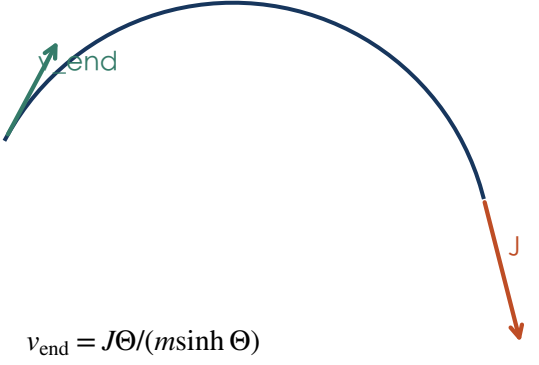
绳的弧线与端点切向冲量；Θ为总转角



$$v_{\text{end}} = J\Theta/(m\sinh\Theta)$$